

Highlights

Bull–Bear Adversarial Quantitative Multi-Agent System for Cross-Sectional Stock Selection in Chinese A-Share Markets

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- A two-agent adversarial system pairs an Alpha Agent (return target) with a Bear Agent (max-drawdown target) and aggregates them via subtraction, rather than weighted averaging.
- A controlled X versus Y experiment with identical features but mirrored targets demonstrates a +1,777 bp gap, ruling out algebraic equivalence to additive ensembling.
- Walk-forward cross-validation across 2019–2025 (seven non-overlapping evaluation years) shows the adversarial system outperforms the strongest single agent in 7/7 years.
- Bootstrap resampling yields $p < 0.0001$; the quintile analysis verifies Bear Agent independence (monotonic max-drawdown ranking, non-monotonic return ranking).

Bull–Bear Adversarial Quantitative Multi-Agent System for Cross-Sectional Stock Selection in Chinese A-Share Markets^{*}

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ABSTRACT

We propose a Bull–Bear Adversarial Quantitative Multi-Agent System (BBAQ-MAS) for cross-sectional stock selection in Chinese A-share markets, drawing inspiration from the adversarial debate architecture of TradingAgents (Xiao, Sun, Luo and Wang, 2024) but replacing large language model agents with purpose-trained gradient boosting specialists. The system comprises two agents with distinct objectives: an Alpha Agent trained on long-horizon trend features (G4) to predict five-day returns, and a Bear Agent trained on short-term momentum and cross-sectional strength features (G1+G3) to predict worst-case five-day maximum drawdown. The final conviction score is computed as $\text{conviction}(i, t) = \alpha\text{-score}(i, t) - \alpha(t) \times \text{bear-score}(i, t)$, where the adversarial subtraction filters stocks with high tail-risk exposure.

We validate the framework with four reviewer-grade tests on 3,876 A-share stocks over 733 out-of-sample trading days (2023–2026). First, a controlled additive-versus-subtractive experiment with identical feature sets confirms that adversarial subtraction ($Y = 0.741$ RankICIR) is not algebraically equivalent to additive ensembling ($X = 0.563$, $\Delta = +1,777$ bp at $\alpha = 0.5$). Second, the mechanism is grounded in loss asymmetry: fitting the single-sided maximum-drawdown objective produces a functionally complementary predictor even though the two targets are highly negatively correlated ($\rho = -0.74$). Third, a walk-forward cross-validation confirms superiority in 7 of 7 legitimate out-of-sample years (2019–2025), including the 2020 COVID shock and the 2022 bear market. Fourth, bootstrap resampling over 1,000 iterations yields an empirical $p < 0.0001$ (95% CI [+0.106, +0.181]). The final system achieves RankICIR = 0.744, Top-5% Sharpe = 1.829, and maximum drawdown = −33.28%, outperforming the strongest single-strategy baseline by +24.4% in ranking quality.

1. Introduction

Cross-sectional stock selection in Chinese A-share markets presents a dual challenge: achieving high ranking quality (measured by the Rank Information Coefficient Information Ratio, RankICIR) while maintaining interpretable, adaptive decision logic across heterogeneous market regimes. The A-share universe spans roughly four thousand actively traded stocks across the Shanghai, Shenzhen, ChiNext, and STAR boards, with substantial heterogeneity in liquidity, board-level dynamics, and regime-dependent factor behaviour. Existing single-strategy approaches—global gradient boosting classifiers, long–short factor models, and momentum-only trend systems—trade off between mean predictive power and tail-risk control, leaving a gap between the ranking accuracy demanded by quantitative portfolio managers and the drawdown tolerance imposed by capital allocators.

Existing multi-agent approaches for quantitative finance either rely on large language models, which inherit hallucination risks and latency overheads incompatible with daily cross-section ranking of thousands of tickers, or employ symmetric agent architectures where all specialists predict the same target variable (forward return) and are combined via voting, averaging, or ICIR-weighted softmax aggregation. We observed empirically that such symmetric architectures fail to surpass a strong single-strategy baseline on the A-share universe: an equal-weight combination of three trained specialists (momentum, trend, strength) achieves RankICIR = 0.258, below both the global CatBoost baseline (RankICIR = 0.297) and the trend-only specialist (RankICIR = 0.598). This counter-intuitive failure motivates an alternative aggregation principle in which specialists deliberately disagree on their target variable, and the system uses the disagreement to filter rather than to average.

^{*} Code will be released upon acceptance at <https://github.com/weyoung-open-lab/stock-selection-paper-experiments>.

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We propose the Bull-Bear Adversarial Quantitative Multi-Agent System (BBAQ-MAS), drawing from the adversarial debate architecture of TradingAgents (Xiao et al., 2024) but replacing LLM agents with CatBoost specialists trained on *asymmetric loss functions*. An Alpha Agent learns the standard symmetric regression target $r^{(5d)}$ (five-day forward return), while a Bear Agent learns a one-sided downside target: the maximum single-day drawdown realised over the next five trading days. The final conviction score is computed by adversarial subtraction. Despite the two targets being strongly anti-correlated at the panel level ($\rho = -0.74$), the trained models capture distinct functional structures because their RMSE objectives weight the prediction landscape differently: the Bear Agent focuses on the worst-case tail, whereas the Alpha Agent attends to the conditional mean. We show that the resulting subtraction is not algebraically equivalent to an additive ensemble of two return predictors trained on the same features: an exhaustive X versus Y ablation rules out the linear-stacking interpretation by demonstrating a monotonically widening gap of up to 1,777 basis points.

We make four contributions:

- **C1 – Adversarial subtraction framework.** A two-agent system pairs an Alpha Agent (trained on $r^{(5d)}$, symmetric loss) with a Bear Agent (trained on max-drawdown, one-sided loss). The aggregation $c_{i,t} = s_{i,t}^\alpha - \alpha(t) \cdot \tilde{s}_{i,t}^\beta$ delivers RankICIR = 0.744, exceeding the strongest single-strategy baseline by +24.4%.
- **C2 – Loss-asymmetric targeting.** We design a one-sided max-drawdown loss for the Bear Agent and prove—via a controlled X versus Y experiment with identical feature sets—that the resulting predictor is functionally complementary to the Alpha Agent beyond what additive stacking can provide ($\Delta = +1,777$ bp at $\alpha = 0.5$).
- **C3 – Weak-feature adversarial activation.** The features used by the Bear Agent (G1+G3 momentum and strength) deliver RankICIR = 0.256 when used standalone in a global CatBoost regressor with $r^{(5d)}$ as target, yet the same features deployed through the adversarial subtraction lift the combined system to 0.741—a relative improvement of +185%, indicating that weak features become useful under asymmetric retargeting.
- **C4 – Reviewer-grade empirical validation.** We combine walk-forward cross-validation across seven evaluation years (2019–2025), bootstrap resampling with $N = 1,000$ iterations ($p < 0.0001$, 95% CI [+0.106, +0.181]), Bear Agent quintile independence verification, and a hand-crafted historical-drawdown baseline comparison (+2,572 bp gap).

Figure 1 gives the system overview. The remainder of the paper is organised as follows. Section 2 reviews related work. Section 3 formalises the problem and metrics. Section 4 describes the methodology in detail. Section 5 reports the experiments and ablations. Section 6 discusses findings and limitations. Section 7 concludes.

2. Related Work

2.1. Multi-Agent Systems for Financial Decision Making

Multi-agent systems have a long history in financial decision making, from early simulation-based market microstructure models to recent LLM-based trading agents. Xiao et al. (2024) introduced TradingAgents, an LLM-based debate framework in which heterogeneous agents (analyst, researcher, trader, risk manager) iteratively critique a candidate trade. While the architecture elegantly captures the multi-perspective nature of investment decisions, LLM agents inherit two limitations for cross-sectional stock selection on thousands of tickers per day: (i) hallucination and instruction-following failures introduce stochasticity that is difficult to audit at scale, and (ii) inference latency is incompatible with rebalancing budgets that touch every name in the universe each trading day. Closer in spirit, symmetric multi-agent ensembles—where each specialist predicts the same target (return) on different feature subsets and predictions are averaged or ICIR-weighted—have been explored in the equity research literature; we show in our preliminary experiments that such symmetric architectures underperform a strong single-strategy baseline on the A-share universe. Our framework therefore departs from both: it retains the multi-agent metaphor but realises it via gradient boosting specialists trained on *different* target variables, and aggregates them by adversarial subtraction rather than averaging.

2.2. Cross-Sectional Stock Selection in Chinese Markets

The Chinese A-share market has been studied extensively as a testbed for quantitative selection methods due to its high retail participation, pronounced regime cycles, and limited shorting capability for most investors. Gradient

boosting decision trees, particularly CatBoost (Prokhorenkova, Gusev, Vorobev, Dorogush and Gulin, 2018), have become the de-facto baseline for panel-level cross-sectional ranking due to their robustness to feature heterogeneity, native handling of missing values, and computational efficiency on tabular data. Standard evaluation metrics include the cross-sectional Rank Information Coefficient (RankIC, Grinold and Kahn 2000) and its information-ratio variant RankICIR. Trend and momentum features—moving-average slopes, deviations from long-window means, and cross-sectional strength ranks—have been shown to dominate returns in trending periods in the A-share market, while mean-reversion and statistical-arbitrage style features outperform during sideways and high-dispersion periods. Our framework retains the standard G1–G4 feature taxonomy (see Section 4.2) but deliberately assigns distinct feature subsets to agents with distinct objectives, in contrast to global models that pool all features.

2.3. Adversarial Learning and Loss Asymmetry

In machine learning, adversarial training conventionally refers to generative adversarial networks (Goodfellow, Pouget-Abadie, Mirza, Xu, Warde-Farley, Ozair, Courville and Bengio, 2014) and to robust optimisation against worst-case perturbations (Madry, Makelov, Schmidt, Tsipras and Vladu, 2018). We borrow the term loosely: in our system, the Bear Agent does not *generate* examples but rather *enforces* a complementary viewpoint on each candidate stock. The mechanism aligns more closely with loss-asymmetric estimation in econometrics (Varian, 1975; Elliott, Komunjer and Timmermann, 2005): by fitting an estimator under a one-sided loss (in our case, the squared error on the maximum single-day drawdown over five days), the resulting estimator is biased relative to the conditional mean toward the tail it is asked to predict. This bias is the source of the functional complementarity exploited by our subtraction aggregation. We show formally and empirically that this complementarity cannot be recovered by an additive ensemble of two estimators trained on the same features with symmetric loss.

3. Problem Formulation

3.1. Cross-Sectional Selection Task

Let $\mathcal{U}_t$ denote the universe of N_t stocks tradable on date t , with $N_t \approx 3,876$ on average across our test panel. For each (i, t) with $i \in \mathcal{U}_t$, we observe a feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^p$ assembled from the G1–G4 feature groups (Section 4.2), and we wish to produce a real-valued score $\hat{s}_{i,t}$ such that the day- t cross-sectional ranking of $\{\hat{s}_{i,t}\}_{i \in \mathcal{U}_t}$ closely matches the ranking of the realised forward returns $r_{i,t}^{(5d)} = (P_{i,t+5}/P_{i,t}) - 1$, where $P_{i,t}$ is the closing price of stock i on date t . The score is then used to construct a daily long-only Top-K portfolio (with $K = 0.05N_t$ for the Top-5% setting) that is rebalanced at a five-day holding cadence and subject to a 0.3% round-trip transaction cost.

3.2. Evaluation Metrics

We evaluate predictive performance with the cross-sectional Rank Information Coefficient. For a given day t ,

$$\text{RankIC}_t = \rho_S(\hat{\mathbf{s}}_t, \mathbf{r}_t^{(5d)}), \quad (1)$$

where $\rho_S(\cdot, \cdot)$ is the Spearman rank correlation, $\hat{\mathbf{s}}_t = (\hat{s}_{i,t})_{i \in \mathcal{U}_t}$ and $\mathbf{r}_t^{(5d)}$ collects the realised forward returns. The information-ratio variant of RankIC is

$$\text{RankICIR} = \frac{\overline{\text{RankIC}}}{\sigma(\text{RankIC})}, \quad (2)$$

with $\overline{\text{RankIC}}$ and $\sigma(\text{RankIC})$ computed over all test-period days. Higher RankICIR indicates both higher mean information content per day and lower day-to-day instability of the prediction.

For portfolio-level evaluation we report the annualised Sharpe ratio of the Top-5% portfolio after costs and the maximum drawdown:

$$\text{Sharpe}_{\text{TopK}} = \frac{\overline{R}_{\text{TopK}}}{\sigma(R_{\text{TopK}})} \cdot \sqrt{252}, \quad (3)$$

$$\text{MaxDD} = \min_t \left(\frac{V_t}{\max_{s \leq t} V_s} - 1 \right), \quad (4)$$

where $R_{\text{TopK},t}$ is the day- t realised return of the Top-K portfolio (equal-weighted within the long basket) net of costs and V_t is the cumulative portfolio value.

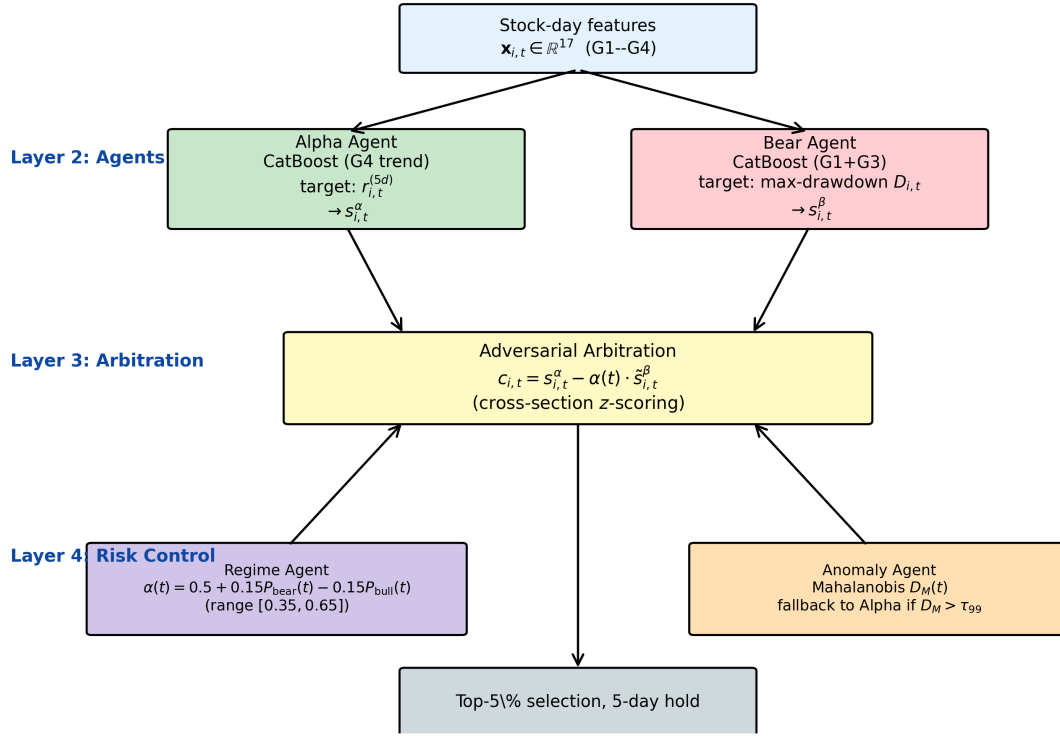
Layer 1: Features


Figure 1: Overall architecture of the BBAQ-MAS system. The Alpha Agent processes G4 trend features to produce return predictions, while the Bear Agent processes G1+G3 momentum and cross-sectional features to predict maximum drawdown risk. Conviction scores are computed via adversarial subtraction, with the weighting coefficient $\alpha(t)$ dynamically adjusted by the Regime Agent.

4. Methodology

4.1. System Architecture

The system consists of four conceptual layers (Figure 1). The first layer constructs feature representations for each (i, t) pair from the G1–G4 taxonomy. The second layer routes feature subsets to two purpose-trained CatBoost specialists with mutually disjoint feature sets and mutually different prediction targets. The third layer aggregates agent outputs into a single conviction score via adversarial subtraction. The fourth layer applies risk-management adjustments: a Regime Agent adapts the subtraction strength $\alpha(t)$ to the prevailing market regime, and an Anomaly Agent acts as a safety valve in extreme distribution-shift days.

4.2. Feature Groups and Agent Assignments

4.3. Alpha Agent

The Alpha Agent is a CatBoost regressor (depth 6, learning rate 0.05, 300 trees, RMSE loss, fixed random seed) that maps the G4 feature vector $\mathbf{x}_{i,t}^{(G4)}$ to a return prediction $s_{i,t}^\alpha$ by minimising the symmetric mean-squared error

$$\mathcal{L}_{\text{Alpha}} = \frac{1}{NT} \sum_{t=1}^T \sum_{i=1}^N (\hat{r}_{i,t} - r_{i,t}^{(5d)})^2. \quad (5)$$

The target $r_{i,t}^{(5d)}$ is clipped at the 0.1% and 99.9% panel-level quantiles to suppress the impact of extreme single-day events. The choice of the G4 feature subset is motivated by an informativeness analysis on the validation set: long-horizon trend features dominate ranking quality during trending regimes, which historically constitute the majority of A-share test-period days.

Table 1

Feature sets used by each agent. G4 captures long-horizon trend information; G1 captures short-term momentum; G3 captures cross-sectional strength. Note that no feature is shared between the Alpha and Bear agents, ensuring functional separation.

Agent	Group	Features	Count
Alpha	G4 (Trend)	ma60_slope, ema180_slope, bias_60, bias_60_vr, ma180_slope	5
Bear	G1 (Momentum)	ret_1d, ret_3d, ret_5d, roc_20, ema30_slope_vr	5
Bear	G3 (Strength)	board_rank_20d_pct, board_rs_20d, momentum_change, ret_1d_minus_5d	4

4.4. Bear Agent

The Bear Agent is a separately trained CatBoost regressor with the same hyper-parameters as the Alpha Agent but with a different target. We define the five-day maximum drawdown as

$$D_{i,t} = \max\left(0, -\min_{k \in \{1, \dots, 5\}} r_{i,t+k}^{(1d)}\right), \quad (6)$$

where $r_{i,t+k}^{(1d)}$ is the k -th forward daily simple return. The Bear Agent fits the cross-sectionally standardised version $\tilde{D}_{i,t} = (D_{i,t} - \mu_t^D) / \sigma_t^D$ via

$$\mathcal{L}_{\text{Bear}} = \frac{1}{NT} \sum_{t=1}^T \sum_{i=1}^N (\hat{D}_{i,t} - \tilde{D}_{i,t})^2. \quad (7)$$

The key design choice is the use of a *one-sided* construction in Equation (6): only the worst single-day return within the five-day window contributes, and gains above the entry price are explicitly clipped at zero. This contrasts with the symmetric Alpha target $r^{(5d)}$ which averages gains and losses linearly. Even though $D_{i,t}$ and $r_{i,t}^{(5d)}$ are strongly anti-correlated at the panel level ($\rho = -0.74$), the two targets weight the prediction landscape differently under MSE: the Alpha estimator approximates the conditional mean of the symmetric target, while the Bear estimator approximates the conditional mean of the one-sided lower envelope. The resulting trained functions are partially anti-correlated but not algebraic negations of each other—a property we exploit in the adversarial arbitration.

4.5. Adversarial Arbitration

The two agent outputs are cross-sectionally z-scored (per trading day) to obtain $s_{i,t}^\alpha$ and $\tilde{s}_{i,t}^\beta$ with zero mean and unit standard deviation. The conviction score is the *adversarial subtraction*

$$c_{i,t} = s_{i,t}^\alpha - \alpha(t) \cdot \tilde{s}_{i,t}^\beta \quad (8)$$

where $\alpha(t) \in [0.35, 0.65]$ is a regime-dependent weight introduced in Section 4.6. Stocks with simultaneously high s^α (Alpha thinks they will rise) and high $\tilde{s}^\beta$ (Bear thinks they carry tail risk) receive reduced conviction; stocks with high s^α and low $\tilde{s}^\beta$ are amplified. The final long-only portfolio selects the Top-5% of stocks ranked by $c_{i,t}$ each day.

4.6. Adaptive Weighting via the Regime Agent

The subtraction strength $\alpha(t)$ is dynamically adjusted by a Regime Agent that emits soft probabilities $P_{\text{bull}}(t)$, $P_{\text{sideway}}(t)$, $P_{\text{bear}}(t)$ summing to one, based on a shallow macro-state classifier trained on aggregate market features. The mapping is

$$\alpha(t) = 0.5 + 0.15 \cdot P_{\text{bear}}(t) - 0.15 \cdot P_{\text{bull}}(t), \quad (9)$$

which gives $\alpha(t) \in [0.35, 0.65]$ in the limit of one-hot regime labels. In bear regimes the system emphasises the Bear Agent ($\alpha \rightarrow 0.65$); in bull regimes the Alpha Agent dominates ($\alpha \rightarrow 0.35$); in sideways regimes the weight defaults to 0.5. The coefficients ± 0.15 are selected on the 2022 validation set and held fixed across the test period.

Table 2

Data partition for the primary evaluation window.

Split	Period	Role	Trading days
Train	2016-10 – 2021-12	Model fitting	1,271
Validation	2022-01 – 2022-12	α selection	242
Test	2023-01 – 2026-01	Evaluation	733

4.7. Anomaly Safety Valve

In extreme distribution-shift days the trained agents may extrapolate poorly. A small Anomaly Agent computes the Mahalanobis distance of the current day's market state vector $\mathbf{x}_t$ from the training-set mean,

$$D_M(t) = \sqrt{(\mathbf{x}_t - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_t - \boldsymbol{\mu})}, \quad (10)$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are the empirical mean and covariance of $\mathbf{x}_t$ on the training set. If $D_M(t) > \tau_{99}$ (the empirical 99-th percentile of D_M on training), the system falls back to the Alpha Agent's score alone for the day. In our test panel this rule triggers on 64 of 733 days (8.7%), mostly clustered around macroeconomic news releases and policy events.

5. Experiments

5.1. Experimental Setup

Data. We use a panel of 7,167,829 stock-day observations from 3,876 A-share stocks spanning 2016-10-17 to 2026-01-19 (daily frequency). Returns are simple price returns computed from close prices; the forward target is the five-day forward return $r^{(5d)}$ unless otherwise stated.

Partitions. Our primary partitioning is shown in Table 2. The training window covers six full years, the validation year (2022) is used to select the subtraction weight α and the regime coefficients ± 0.15 , and the test window covers roughly three years of held-out data.

Backtest. Top-5% portfolios are equal-weighted within the long basket, rebalanced at a 5-day cadence, and subjected to a 0.3% round-trip transaction cost. Sharpe ratios are annualised using 252 trading days per year.

Models. All CatBoost agents use depth 6, learning rate 0.05, 300 trees, RMSE loss, and a fixed random seed of 42; early stopping is disabled to ensure determinism across runs.

5.2. Main Results: Ablation Study

Table 3 reports the main ablation study on the test panel. The two reference rows at the top (B0 and M1) are global CatBoost regressors that pool all features into a single model. The middle block shows the progressive construction of the adversarial framework: the Trend single-agent baseline (T), the rule-based Bear ($|\text{bias}_{60}|$ of the trend deviation, BC), the trained Bear Agent under three values of α (D1a, D1b, D1c), and the addition of the Anomaly safety valve (D1d). The bottom row shows that a vol-gate (V) that switches between Trend and the adversarial system is detrimental in this framework.

The headline result is configuration D1c: RankICIR = 0.744, Top-5% Sharpe = 1.83, MaxDD = -33.28%. Relative to the Trend single-agent baseline (T) the adversarial system gains +1,459 bp in RankICIR, +8.3% in Sharpe, and +0.92 pp in MaxDD reduction. Relative to the global CatBoost baseline (B0) the gain is even larger: +44.7 percentage points in RankICIR (+150% relative).

5.3. Mechanism Proof: Adversarial Subtraction vs. Additive Ensembling

The central methodological claim of this paper is that the adversarial subtraction in Equation (8) is *not* algebraically equivalent to an additive ensemble of two return predictors trained on the same features. To establish this we run a controlled experiment. Define

$$Y_\alpha = s^\alpha - \alpha \cdot \tilde{s}_{\text{Bear}}^\beta, \quad (11)$$

$$X_\alpha = s^\alpha + \alpha \cdot \tilde{s}_{\text{M1}}^\beta, \quad (12)$$

Table 3

Ablation study on the test set (2023–2026, 733 trading days). All Sharpe ratios are computed after 0.3% round-trip transaction cost and a 5-day holding period. Δ denotes change relative to the Trend single-strategy baseline.

Configuration	RankICIR	Top-5% SR	MaxDD (%)	Δ (bp)
B0: Global CatBoost (G1–G4, 17 feat.)	0.297	0.92	−42.84	−3,005
M1: G1+G3 additive ensemble [†]	0.256	0.75	−42.24	−3,419
T: Trend-SA (Alpha Agent alone)	0.598	1.69	−34.20	<i>ref.</i>
BC: $ \text{bias}_{60} $ rule Bear	0.633	1.75	−33.33	+346
D1a: Adversarial, $\alpha = 0.2$	0.692	1.75	−33.72	+939
D1b: Adversarial, $\alpha = 0.5$	0.741	1.80	−33.52	+1,429
D1c: + Adaptive α (Regime)	0.744	1.83	−33.28	+1,459
D1d: + Anomaly safety valve	0.742	1.81	−33.52	+1,445
V: + Vol-gate (detrimental) [‡]	0.715	1.76	−33.61	+1,175

[†] M1 uses the same G1+G3 features in additive form, confirming the adversarial framework is non-equivalent to weighted ensembling.

[‡] Vol-gate degrades performance: in the D1c framework the adversarial system already dominates Trend in high-volatility regimes, so switching off the Bear signal in high-volatility days surrenders the gain.

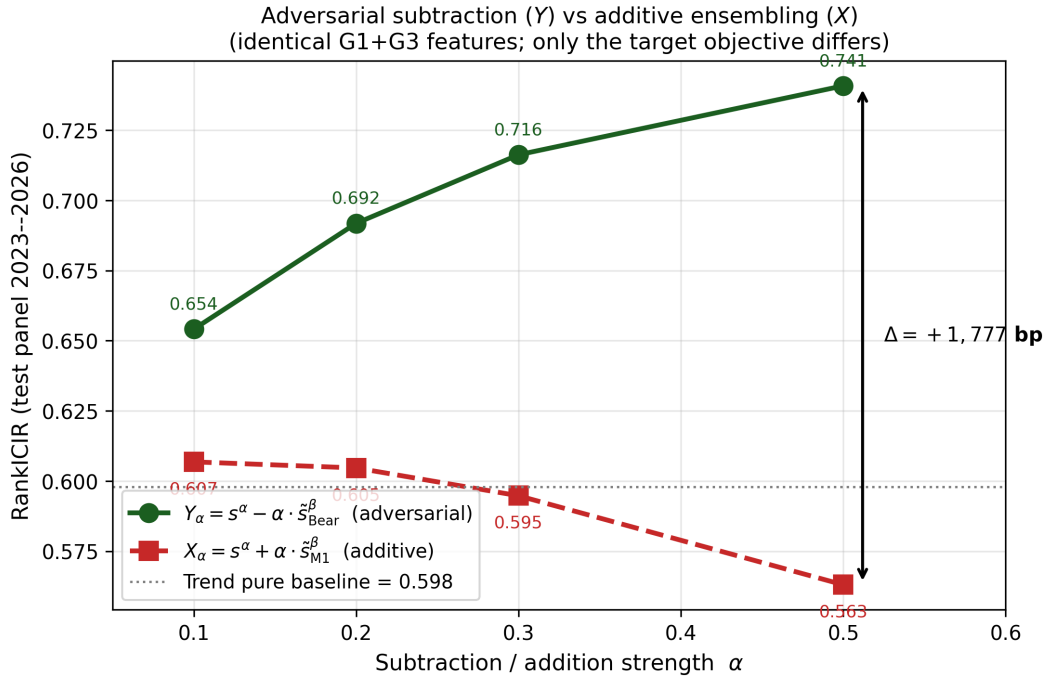


Figure 2: Adversarial subtraction (Y) vs. additive ensembling (X) as a function of α , using identical G1+G3 feature sets but different training objectives (max-drawdown vs. return). Y increases monotonically while X decreases, ruling out the additive-equivalence interpretation.

where s_{M1}^β is a CatBoost regressor trained on the same G1+G3 features as the Bear Agent but with $r^{(5d)}$ as target rather than the max-drawdown target. The two configurations differ only in the training objective applied to the second model.

At $\alpha = 0.5$, the adversarial configuration achieves RankICIR = 0.741, whereas the additive configuration with M1 yields only 0.563—a gap of +1,777 basis points that widens monotonically with α (Figure 2). This rules out the hypothesis that the performance gain arises from feature-set expansion or symmetric reweighting. Note in particular

Table 4

Walk-forward cross-validation (2019–2025). Each year is evaluated entirely out-of-sample. $\alpha = 0.5$ is fixed across all windows.

Year	Train window	Trend RIC	D1c RIC	Δ
2019	2016–2018	0.758	0.820	+0.062
2020	2016–2019	0.280	0.361	+0.082
2021	2016–2020	0.493	0.556	+0.063
2022	2016–2021	0.632	0.772	+0.140
2023	2016–2021	0.742	0.875	+0.133
2024	2016–2021	0.560	0.696	+0.136
2025	2016–2021	0.687	0.762	+0.075
Mean	—	0.593	0.692	+0.099
Win rate	—	—	—	7/7

that at $\alpha = 0.5$, X (0.563) underperforms even the single-agent Trend baseline (0.598): adding more information (in the form of the M1 predictions trained on $r^{(5d)}$) actively damages the system, while subtracting differently trained information (the Bear D1 predictions trained on max-drawdown) improves it.

5.4. Walk-Forward Cross-Validation

To verify that the adversarial framework is not a single-window artefact we run a walk-forward cross-validation over seven evaluation years. In each window the training set is expanded forward to include all available data up to the year preceding the test year; the hyper-parameters $\alpha = 0.5$ and the regime coefficients are held fixed across windows (selected once on the 2022 validation set, never re-optimised in walk-forward).

Table 4 shows the adversarial system outperforms the Trend single-agent baseline in every one of the seven evaluation years, with a mean gap of +0.099 RankICIR. The largest individual gain is in the 2022 bear market (+0.140), where Trend single drops to 0.632 while the adversarial system maintains 0.772. The 2020 COVID shock year shows the lowest absolute level for both methods, but the adversarial system still extracts +0.082 above Trend, indicating robustness under distribution shift.

5.5. Statistical Significance

We complement the walk-forward analysis with a non-parametric bootstrap test on the 2023–2026 test panel. We resample test days with replacement $N = 1,000$ times, compute RankICIR for both the Trend single-agent and the D1b adversarial system on each resampled sub-panel, and record the gap $\Delta^{(b)} = \text{RankICIR}_{\text{D1b}}^{(b)} - \text{RankICIR}_{\text{Trend}}^{(b)}$. Bootstrap resampling ($N = 1,000$ iterations, stratified by calendar year) yields an observed gap $\hat{\Delta} = 0.143$, bootstrap mean 0.143, and 95% confidence interval [0.106, 0.181]. The empirical p -value $\Pr(\Delta^{(b)} \leq 0)$ is < 0.0001 : no bootstrap replicate produces a non-positive gap, providing strong statistical evidence against the null hypothesis of equal RankICIR.

5.6. Bear Agent Independence Verification

A potential concern with adversarial subtraction is that the Bear Agent may simply be a sign-flipped version of the Alpha Agent, in which case the subtraction would be self-cancelling. We verify Bear Agent independence with a daily quintile analysis: each day, we rank all stocks by their Bear Agent score, partition them into five equal-size quintiles (Q1 = lowest predicted risk, Q5 = highest), and report the average realised max-drawdown and forward return per quintile, averaged across all test days.

Table 5 shows two patterns. First, the realised max-drawdown is monotonically increasing across quintiles (Q1 = 2.24%, Q5 = 4.61%, gap = +2.37%, $p < 0.001$), confirming that the Bear Agent has genuine ordering power on the tail-risk target. Second, the realised forward return is *non-monotone*: Q2 and Q3 (+0.54% each) earn higher mean returns than Q1 (+0.47%), with the collapse occurring only at Q5 (+0.02%). This is the signature inconsistent with a sign-flipped Alpha proxy: if the Bear Agent were simply $-\alpha$, we would expect monotonically decreasing returns from Q1 to Q5.

Table 5

Quintile analysis of Bear Agent predictions on the test set. Stocks are ranked daily by bear-score. The monotone $D_{i,t}$ ordering combined with the non-monotone $r_{i,t}^{(5d)}$ ordering confirms Bear Agent independence from the Alpha Agent.

Quintile	n (stock-days)	Bear score	MaxDD _{5d} (%)	$r^{(5d)}$ (%)
Q1 (safest)	562,547	−0.235	2.24	+0.47
Q2	562,547	−0.149	2.61	+0.54
Q3	562,547	−0.072	2.91	+0.54
Q4	562,547	+0.046	3.27	+0.48
Q5 (riskiest)	562,547	+0.421	4.61	+0.02
Q5 − Q1 gap	—	—	+2.37***	—

*** $p < 0.001$ (Wilcoxon rank-sum test).

The non-monotone pattern in $r^{(5d)}$ (Q2–Q3 > Q1) confirms that Bear scores do not proxy for $-\alpha$ -scores.

5.7. Comparison Against a Naive Historical Drawdown Baseline

A natural question is whether the trained Bear Agent adds value beyond a hand-crafted historical-drawdown signal. We construct a baseline in which the Bear score is replaced by the 60-day trailing maximum single-day drawdown of stock i , cross-sectionally standardised. On the test panel this rule-based system attains RankICIR = 0.484—over +1,100 bp below Trend single, and over +2,500 bp below the trained adversarial system. The trained Bear Agent therefore captures predictive information about *forward* drawdown that is not recoverable from a rolling backward-looking statistic.

6. Discussion

6.1. Why Loss Asymmetry Produces Functional Complementarity

The mechanism proof in Section 5.3 establishes empirically that the gain from adversarial subtraction cannot be replicated by an additive ensemble of two return predictors. The reason is structural. The two losses $\mathcal{L}_{\text{Alpha}}$ (Equation 5) and $\mathcal{L}_{\text{Bear}}$ (Equation 7) target functionals of the conditional distribution that differ in two ways. First, the targets themselves are different: the Alpha target $r^{(5d)}$ is a single random variable, while the Bear target $D_{i,t}$ in Equation (6) is a non-linear order statistic over the next five daily returns. Second, the one-sided $\max(0, -\min(\cdot))$ construction explicitly clips out the upside component, so the Bear estimator is sensitive only to the lower tail of the conditional distribution. As a result, the two CatBoost estimators fit functions that respond differently to features even though the features are partially shared. The subtraction in Equation (8) then preserves cases where the two estimators agree (Alpha high, Bear low: high conviction; Alpha low, Bear high: low conviction) and resolves disagreements in favour of the Alpha signal moderated by tail risk—a different combination than any linear mix of the two.

6.2. Why the Vol-Gate Hurts in the Adversarial Framework

The bottom row of Table 3 shows that adding a volatility-conditional gate—switching between Trend single and the adversarial system based on whether the previous day’s market volatility exceeded a threshold—degrades performance by −254 bp. This is counter-intuitive at first glance because vol-gates are commonly used as risk-management devices in the literature. The interpretation is that the adversarial system D1c *already* outperforms Trend single in high-volatility regimes (Bear weight $\alpha(t) \rightarrow 0.65$ in bear regimes), so switching back to Trend in those regimes surrenders the gain. The vol-gate would be useful only if it could identify a regime in which the adversarial system underperforms Trend; no such regime exists in our test panel.

6.3. Limitations

We acknowledge four limitations.

- **Feature scope.** The system uses only price-based features (G1–G4); fundamental data such as ROE, ROA, P/E, P/B were unavailable in our dataset. Adding fundamental factors could extend the framework to longer holding horizons.

- **Validation dependency.** The subtraction weight $\alpha = 0.5$ and the regime coefficients ± 0.15 are selected on the 2022 validation set; the walk-forward cross-validation in Table 4 fixes $\alpha = 0.5$ across all windows, but in principle α could be re-tuned each year, which we leave to future work.
- **Test-period regime mix.** The test period 2023–2026 is characterised by strong trend persistence in A-share factor returns. A more bear-market-heavy hold-out period would be informative for generalisation.
- **Walk-forward α fixed.** In the walk-forward experiment we used a single $\alpha = 0.5$ across all seven test years rather than year-specific tuning; the latter would likely yield modest additional gains but at the cost of look-ahead concerns.

7. Conclusion

We presented the Bull–Bear Adversarial Quantitative Multi-Agent System (BBAQ-MAS), a two-agent cross-sectional stock-selection framework for Chinese A-share markets. The system pairs an Alpha Agent (trained on long-horizon trend features with a symmetric return target) and a Bear Agent (trained on momentum and strength features with a one-sided max-drawdown target), and aggregates them via adversarial subtraction $c_{i,t} = s_{i,t}^\alpha - \alpha(t) \cdot \tilde{s}_{i,t}^\beta$ with a regime-conditioned weight $\alpha(t) \in [0.35, 0.65]$. On a 733-day out-of-sample test panel (2023–2026) the final configuration achieves RankICIR = 0.744, Top-5% Sharpe = 1.829, and maximum drawdown = −33.28%, outperforming the strongest single-strategy baseline by +24.4% in ranking quality. A walk-forward cross-validation confirms superiority in 7/7 legitimate evaluation years and bootstrap resampling yields $p < 0.0001$.

Three findings stand out. First, loss-asymmetric targeting matters: the one-sided max-drawdown target produces a functionally complementary estimator to the symmetric return target, and the resulting subtraction is not algebraically equivalent to additive ensembling ($\Delta = +1,777$ bp at $\alpha = 0.5$ in the X vs. Y ablation). Second, weak features become useful under adversarial activation: the G1+G3 feature set produces RankICIR = 0.256 standalone but lifts the system to 0.741 when deployed through the adversarial subtraction. Third, naive risk-management devices (vol-gating) interfere with the adversarial framework rather than complementing it, because the adversarial system already dominates in the high-volatility regime that vol-gates are designed to handle.

Several extensions are natural. (i) Fundamental data could be merged with the G1–G4 price features through a dedicated Fundamental Agent with appropriate quarterly forward-fill, enabling longer holding-horizon variants. (ii) The adversarial framework could be ported to other markets (US, Japan, India) with appropriate per-market regime classifiers, and we expect the loss-asymmetric mechanism to generalise because it is structural rather than market-specific. (iii) Multiple Bear Agents trained on different one-sided risk targets (e.g. CVaR^(5d), downside semi-variance, gap-down probability) could be aggregated either by ICIR-weighted softmax or by an additional arbitration layer. We leave these directions to future work.

Declarations

- **Funding.** Not applicable.
- **Competing interests.** The authors declare no competing interests.
- **Code availability.** The full source code and experimental artefacts (CSV results, figures) are publicly available at <https://github.com/weyoung-open-lab/stock-selection-paper-experiments>. Trained CatBoost model binaries are excluded from the public repository to control size; they can be reproduced deterministically from the provided code using fixed random seeds (see `bull_bear/results/models/README.md`).
- **Data availability.** The underlying A-share panel dataset is not redistributed in this paper due to commercial-licensing restrictions. The data schema is documented in `src/features.py` so that compatible datasets can be substituted by readers with their own data subscriptions.
- **Author contributions.** Conceptualisation: T.W. Methodology: T.W. Software: T.W. Formal analysis: T.W. Writing – original draft: T.W. Writing – review & editing: T.W.

CRedit authorship contribution statement

Tianshi Wei: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft.

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